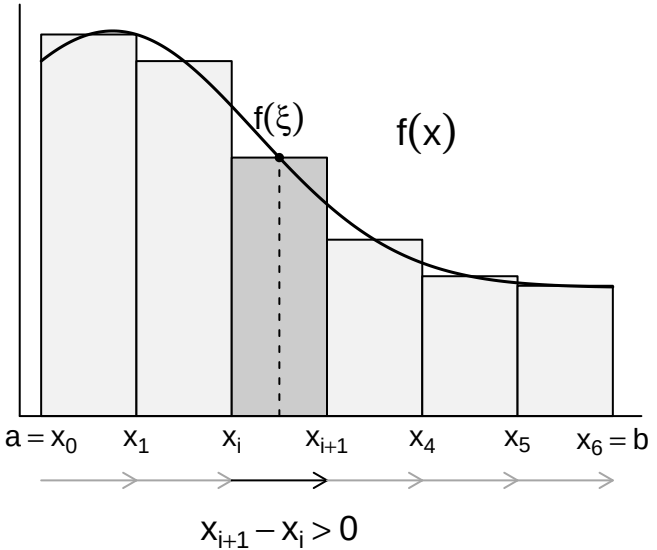
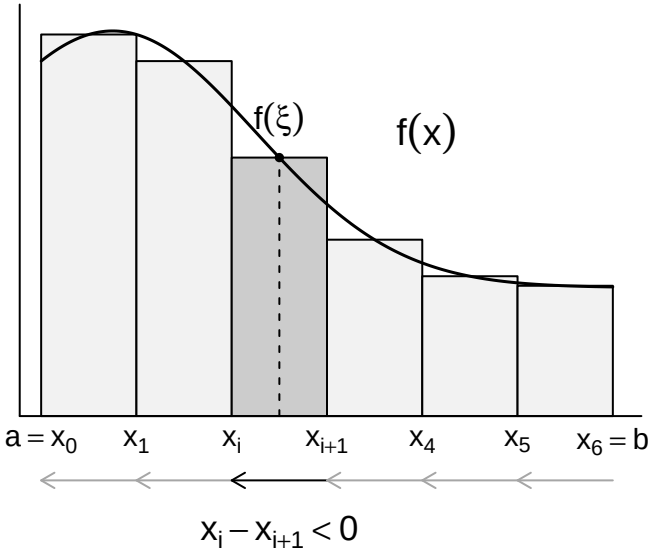
$$\int_a^b f(x)dx$$



$$x_{i+1} - x_i > 0$$

$$\int_b^a f(x)dx$$



$$x_i - x_{i+1} < 0$$